

Fake News Detection Approach Using Hybrid Deep Learning Framework

Maria Vasiliki Sanida¹, Theodora Sanida², Argyrios Sideris², Michael Dossis² and Minas Dasygenis²

¹Department of Digital Systems, University of Piraeus 18534, Piraeus, Greece

²Department of Electrical & Computer Engineering, University of Western Macedonia, 50131, Kozani, Greece
sanidasilia@gmail.com, thsanida@uowm.gr, asideris@uowm.gr, mdossis@uowm.gr, mdasyg@ieee.org

Abstract—The rapid expansion of social media has facilitated individuals' widespread distribution of information, frequently without sufficient verification or filtering mechanisms. This has exacerbated the problem of misinformation, raising significant concerns about the harmful impacts of fake news on communities. To counteract this issue, a focused effort has been made to develop automated detection systems that harness artificial intelligence (AI) and machine learning (ML) capabilities. Given the success of deep learning (DL) techniques in managing complex natural language processing (NLP) tasks, these methods offer a promising avenue for detecting fake news. This study introduces an innovative DL framework that effectively combines convolutional neural networks (CNNs) and recurrent neural networks (RNNs) to detect fake news. The proposed approach attains 99.19% accuracy and significantly superior detection outcomes compared to existing methods, highlighting its potential to significantly enhance the reliability of news verification processes and mitigate the spread of misinformation.

Index Terms—Fake news detection; Misinformation; Information dissemination; Deep learning; Optimization algorithms.

I. INTRODUCTION

The rapid advancements in artificial intelligence (AI) have spurred many experiments to manage challenges previously unexplored within computer science. Fake news detection exemplifies one such challenge. As news media accessibility has increased, any significant global event promptly triggers various news outlets to enhance the appeal of their reports. This strategy aims to maximize the reach of their content across the worldwide web. Consequently, news spreads swiftly to millions or billions of individuals via numerous platforms, including television channels, articles, websites, and social networking sites. This widespread dissemination poses unique challenges and opportunities for developing effective detection mechanisms in the digital age [1].

In addition to several well-established and reputable international news organizations that disseminate information to the public, a plethora of smaller, less reliable news sources exist. Furthermore, on famous social networking and media frameworks, individuals from anywhere in the world can post and distribute any statement or set of statements, often propagating fake news to achieve various potentially fraudulent or illegal objectives. Additionally, certain official news aggregators might deliberately circulate false information to increase their popularity, achieve political aims, or generate

revenue, exacerbating the problem. Another contributing factor to the stretch of fake news is scheduled campaigns designed to discredit or damage the reputation of specific products, societies, companies, or groups for social, political, or financial motives [2].

Deep learning techniques [3] have appeared as a powerful mechanism in the battle against fake news, leveraging algorithms to parse and understand the nuances of human language. These techniques employ layered neural networks that mimic human cognitive processes, enabling them to analyze vast amounts of data for patterns that might indicate false information. Training these models on large datasets of verified true and false news stories teaches them to differentiate between factual reporting and misinformation. This capability is precious given the subtlety and complexity of many deceptive news articles, which may contain a mix of truth and falsehoods intricately woven together. As these technologies continue to advance, they offer the potential to significantly enhance the accuracy and speed of fake news detection, providing a critical tool to preserve the integrity of information on digital platforms [4].

In overview, the contributions of the study is as follows:

- We develop a robust and highly accurate resolution fake news detection approach, leveraging the strengths of both convolutional neural networks (CNNs) and recurrent neural networks (RNNs). This framework capitalizes on CNN's ability to process and analyze patterns and the RNN's strength in managing sequential data in fake news articles. This methodology not only enhances the precision of fake news detection but also improves the model's ability to adapt to the dynamic and continuously changing tactics employed by sources of misinformation.
- In order to optimize the output of our model, we conduct a comprehensive evaluation using a variety of metrics. This assessment process ensures the model's accuracy, reliability, and robustness across diverse scenarios. By analyzing metrics such as recall, precision, and F1-score, we can identify strengths and weaknesses in the model's performance, mainly how it handles different types of fake news scope.

The remaining parts of this study have the following organizational structure: We provide fake news analysis investiga-

tions in Part II, and Part III analyses our approach's constituent components and methods. Part IV demonstrates the study's outcomes, and Part V summarises our study outcomes and future analysis.

II. RELATED WORK

Pursuing robust news verification mechanisms and debunking false information on the web constitutes a multi-faceted research endeavour to craft highly effective solutions. Despite its popularity, this field of inquiry remains nascent, with significant scope for exploration and discovery. Although there has been an increase in the number of research focused on the examination and analysis of false news, the lack of a standardised and widely acknowledged solution highlights the intricate nature of the problem.

The work by Ahmed et al. [5] serves as a comprehensive examination of various ML methodologies applied to the ISOT dataset; included in their study were the Support Vector Machine (SVM) and its variant, the Linear Support Vector Machine (LSVM), which are esteemed for their capacity to discover the optimal hyperplane in a high-dimensional space that most suitably segregates the data into distinct categories. Furthermore, the study explored the K-Nearest Neighbor (KNN) method and the Stochastic Gradient Descent (SGD) method, an optimization technique capable of managing large datasets. The decision-making process of the Decision Tree (DT) classifier was also surveyed. The Logistic Regression (LR) method was also examined for its proficiency in estimating category membership probabilities based on independent predictors. The higher accuracy is attained by LSVM, which has a 92% accuracy rate.

The authors proposed the WOA-Xgbtree method in [6]. This algorithm combines the Extreme Gradient Boosting Tree (xgbTree) algorithm with the Whale Optimisation method (WOA) to detect fake news items based on extracted features. The outcomes demonstrate that the suggested approach gained accuracy in 91.86% of the ISOT dataset. Also, in [7], they proposed a method via Graph-based Markov chains. The results indicate that the presented technique earned accuracy in 97.70% of the ISOT dataset. Additionally, in [8], the authors presented automating fake news detection utilizing Probabilistic Principal Component Analysis (PPCA) and levy flight-based Long Short-Term Memory (LSTM). The results display that the proposed method reached accuracy in 98.50% of the ISOT dataset.

Unlike the prior authors, our study introduces a resilient and highly precise approach to fake news detection, employing a synergistic integration of CNNs and RNNs. This advanced methodological framework capitalizes on the CNNs' capability for extracting salient features through their robust hierarchical pattern recognition and the RNNs' adeptness at processing sequential data, which is essential for capturing the temporal dynamics of language. The amalgamation of these neural network architectures not only amplifies the accuracy of discerning false news narratives but also confers upon the model an enhanced capacity for adjusting to the ever-evolving

strategies utilized by purveyors of disinformation. This adaptability is critical in an environment where misinformation tactics are not static but continuously refined to circumvent detection. Through this innovative approach, we significantly elevate the standard for automated fake news identification, ensuring robust performance even as the landscape of digital misinformation transforms.

III. MATERIALS AND METHODS

A. Fake news dataset

The ISOT dataset, a robust collection used extensively in fake news detection research, includes 45,000 news articles split between real and fake categories. The real articles within the dataset were sourced from Reuters, a well-regarded news organization known for its comprehensive and accurate reporting. Conversely, the fake articles were compiled from various outlets that were identified as unreliable sources by entities such as Wikipedia and fact-checking organizations like Politifact. This dataset includes each article's full text and provides additional metadata such as titles, publication dates, and topics, enriching the context available for analysis. The articles primarily focus on politics and world news, with the content dating from 2016 to 2017. This timeframe is particularly significant as it encompasses numerous globally impactful events, offering a dynamic range of material for testing and refining fake news detection algorithms [5]. The allocation by category and topics is shown in Table I.

TABLE I
THE DISTRIBUTION OF THE ISOT DATASET.

Category	Number of articles	News	Total number of articles
World news	10145	Real	21417
Politics news	11272		
Government news	1570	Fake	23481
US news	783		
News	9050		
Middle east	778		
Left news	4459		
Politics	6841		

B. The proposed approach for framework

The model architecture in Figure 1 is a neural network tailored for a fake news framework. At the outset, an input layer accepts variable-length sequences where articles may vary in length. Following the input, an embedding layer maps separate tokens to a high-dimensional vector space. This embedding process captures semantic relationships between tokens and projects them into a continuous vector space where neural network layers can process them. The embedded vectors then pass through a convolutional layer to capture local patterns within the embedding space. Convolutional layers are adept at determining spatial hierarchies in data and are equally useful in text for picking out invariant features such as common n-grams or phrases. The convolution is followed by a MaxPooling1D layer, which reduces the data's dimensionality by pooling over time steps, retaining only the most salient features, maximizing the signal and minimizing noise [9].

The sequential nature of the text data is handled next by a bidirectional Long Short-Term Memory (Bi-LSTM) layer. LSTM units are particularly suited for capturing long-range dependencies within sequences, an attribute critical in understanding language's context and structure. The bidirectionality implies that the LSTM reads the data forwards and backwards, integrating context from both directions, thus providing a fuller understanding of the sequence. This combination of convolutional and LSTM layers is repeated, albeit with a convolutional output reduced to 64 dimensions. This pattern suggests a deliberate design to model complex features at multiple scales of the data's structure, abstracting higher-level features from the raw input.

The final layers of the network consist of a dense layer with 32 units headed by a dropout layer. The dense layer is a fully connected neural network layer that integrates the learned features into higher-level representations. The dropout layer is a regularization technique used to prevent overfitting, which is particularly important in DL models that might otherwise learn to replicate the training data similarly closely and fail to generalize to newly unseen data. Finally, the framework has a single-unit dense layer for binary detection tasks, where the single unit represents the probability of the input to one of two categories, real or fake. The output of this final layer would be used to determine the category of the input sequence, thus completing the process of detecting the input text data as either fake or real news. This model represents an advanced integration of CNN and RNN layers for managing complex sequence category tasks, leveraging their respective strengths in feature extraction and sequence modelling to create a nuanced and effective tool for text detection.

C. Approach implementation

The proposed approach is implemented utilizing the sequential model of the Keras DL Python library. The ISOT collection is separated into training/test subsets (80%/20% split) [10]. The approach's training regimen incorporates the adaptive moment estimation (Adam) algorithm optimizer. The binary cross-entropy loss function is employed to quantify the distinction between the predicted probabilities and the actual binary labels of the dataset. The training process is conducted throughout 10 epochs to ensure that the model is exposed to the intricacies of the data multiple times, thus learning to generalize better. Using a batch size of 32 strikes a balance between computational efficiency and the stability of the learning process.

D. Evaluation measurements

The efficacy of the proposed approach for fake news detection was evaluated using four measures [11], as delineated from (1) to (4). The accuracy measure quantifies the fraction of correct predictions across all evaluated instances, offering an overview of the model's general predictive capabilities. Precision is computed as the share of accurate positive predictions out of the whole number of positive predictions made, reflecting the exactitude of the framework in detecting instances

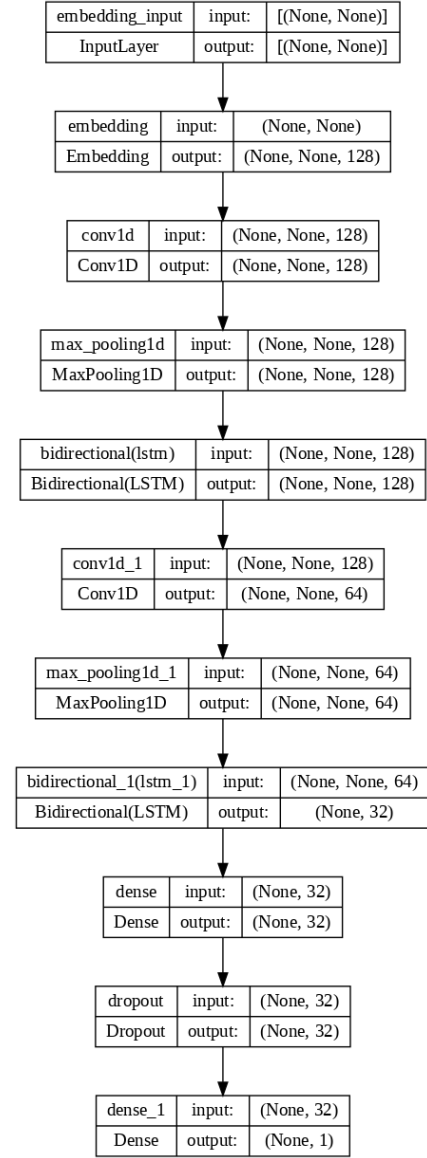


Fig. 1. ISOT model for fake news detection.

as positive. Recall, in turn, is defined as the proportion of correctly detected positive instances relative to the entire pool of actual positive cases, providing insight into the model's sensitivity. Lastly, the F1-Score is derived as the harmonic mean of recall and precision, serving as a single measure that balances both the precision of positive predictions and the completeness with which positive instances are detected.

$$\text{Accuracy} = \frac{APos + ANeg}{APos + WNeg + WPos + ANeg} \star 100\% \quad (1)$$

$$\text{Recall} = \frac{APos}{APos + WNeg} \star 100\% \quad (2)$$

$$\text{Precision} = \frac{APos}{APos + WPos} \star 100\% \quad (3)$$

$$\text{F1-Score} = \frac{2 \star (\text{Recall} \star \text{Precision})}{(\text{Precision} + \text{Recall})} \star 100\% \quad (4)$$

Equations (1) to (3)'s symbols WPos, APos, ANeg, and WNeg stand for the number of wrong positives, actual positives, actual negatives, and wrong negatives, respectively.

IV. EXPERIMENTAL OUTCOMES

In the classification report in Table II for the fake category, the precision metric is 98.61%, the recall is 99.69%, and the F1-score is 99.15%. In the real category, the precision is 99.72%, recall is 98.73%, and the F1 score is 99.22%. Overall, the model's accuracy is reported at 99.19% across 8980 instances, a testament to the model's robustness in distinguishing between fake and real news. This level of precision and reliability in discerning truth from misinformation is not just statistically significant but also socially critical, considering the ramifications of fake news on public opinion and societal trust.

TABLE II
CLASSIFICATION REPORT FOR FAKE NEWS DETECTION

Category	Precision	Recall	F1-Score	Support
Fake	98.61	99.69	99.15	4258
Real	99.72	98.73	99.22	4722
Accuracy			99.19	8980

The confusion matrix in Figure 2 for news detection offers a detailed portrayal of a category model's performance, highlighting its ability to distinguish between real and fake news. So, in the case of fake news detection, the model has correctly identified a substantial majority of fake articles (4245 out of 4258), underscoring its proficiency in detecting patterns or features commonly associated with misinformation. Similarly, the model has correctly categorized 4662 out of 4722 instances for real news, demonstrating its effectiveness in verifying authentic information. However, the model is not without error; there are 13 instances where real news articles were mislabeled as fake and 60 instances where fake articles were mistaken for real.

V. CONCLUSIONS AND FUTURE WORK

This work has managed the pervasive misinformation proliferating through social media. By leveraging the powers of convolutional and recurrent neural networks, our proposed model has demonstrated exceptional performance, achieving an accuracy rate of 99.19%, which marks a substantial improvement over existing fake news detection methodologies. While the current model excels in accuracy, the complexity of fake news

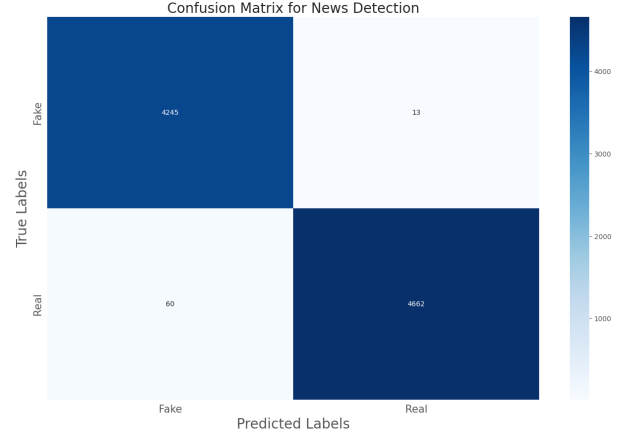


Fig. 2. Confusion matrix for news detection.

propaganda and its evolution call for ongoing adaptation and refinement of detection algorithms. More complicated neural network architectures and transfer learning will be part of our future investigation.

ACKNOWLEDGMENTS

This work was partially supported by the University of Western Macedonia.

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